

Water-NaCl Medium and Tank Modeling Note - Rev A

1. Scope

The benchmark uses a custom water-NaCl medium because a suitable mixture was not available in the baseline Modelica.Media set used by the source study. Intended hierarchy: PartialMedium -> PartialMixtureMedium -> PartialMixtureTwoPhaseMedium -> WaterNaCl, with a one-phase specialization where required.

2. Tank balance equations

```
medium.p = p_ambient;  
V = area*level + V0; m = V*medium.d;  
mXi = m*medium.Xi; U = m*medium.u;  
der(m) = sum(topPorts.m_flow) + sum(ports.m_flow);  
der(U) = sum(H_flow) - p_ambient*der(V);
```

For an incompressible formulation, the $p_{\text{ambient}} \cdot \text{der}(V)$ contribution may be omitted to avoid small artificial temperature changes.

3. Port crossing and numerical robustness

Preferred treatment: asymmetric loss factors with a regularized quadratic flow law. When a port is above liquid level, use a very large outflow loss factor (default 1e5) and small inflow factor (default 0.01). Apply hysteresis based on port diameter around level crossing. State-machine and declarative curve-switch alternatives are historical options only.

4. Evaporator/condensation logic

```
HeatB5.Q_flow = if controller.actuators.T5_Heater then 20000 else 0;  
if Q_up > heatPort.Q_flow then  
  m_con = 0;  
else  
  m_con = -(heatPort.Q_flow - Q_up) /  
    (h_v - h_l + Xi[1]*(cp*dT_dX + dh_l_dX));  
end if;
```

Q_{up} suppresses repeated switching around the boiling point when colder feed enters. K1 is physically separate, while the legacy simulation combines condenser behavior into the B5 evaporator model and exports condensate through an additional fluid port.

5. Composition-dependent properties

Required properties: density $\rho(T,p,w)$, enthalpy $h(T,p,w)$, $cp(T,w)$, saturation pressure $p_{\text{sat}}(T,w)$, and viscosity $\eta(T,w)$. Some source correlations omit numeric coefficients and those coefficients are intentionally absent from this benchmark evidence.

```
eta(T,w) = exp(c1*T^3 + c2*T^2 + c3*T + c4 + c5*w + c6*w^2 + c7*w^3)  
c1=-6.83241E-07 c2=0.000765676 c3=-0.297018665  
c4=-0.299730079 c5=2.065267182 c6=1.546491257 c7=31.57571595
```

6. Known issue

The legacy study reports that the full plant runs with StandardWater but the WaterNaCl configuration can stop when a return pump is scheduled to start. Treat this as a known numerical/modeling issue, not evidence that the physical pump should be omitted.